

KW007: Ultra-Low Power 5.8GHz Microwave Radar Sensor

Motion Detection with Intelligent Doppler Engine

Programming Guide

Version 1.9

March 2026

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General Description:

The **KW007** is a **Doppler radar sensor** designed for ease of use; detection results can be **read from pin** directly, or the device can be configured and monitored via an **I²C interface**.

Once configured, the device operates autonomously, detecting object motion relative to the sensor and updating status flags to indicate **approaching** or **leaving** targets.

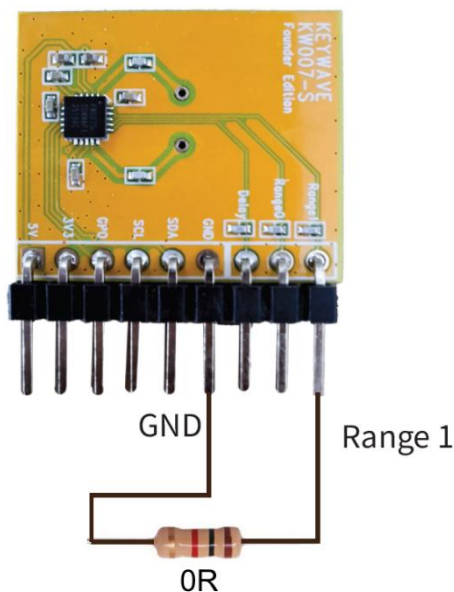
Key Operational Characteristics:

- The internal status flags (Approach/Leaving) are updated at an approximate rate of **8 Hz**.
- Hardware flags can be read directly via **I²C** for low-power **MCU** designs
- **VCO bank calibration** is factory-programmed and must be read during initialization
- Detection range is configurable through **amplifier gain** and **threshold** settings

Applications:

- ◆ **Smart Lighting:** Motion-triggered indoor/outdoor light control.
- ◆ **Touchless Control:** Hands-free faucets, soap dispensers, and automatic lids.
- ◆ **Presence Detection:** Proximity sensing for kiosks, digital signage, and smart displays.
- ◆ **Device Wake-up:** Low-power "wake-on-approach" for IoT devices and security panels.
- ◆ **Home Automation:** Context-aware sensing for smart home ecosystems.

Hardware Note

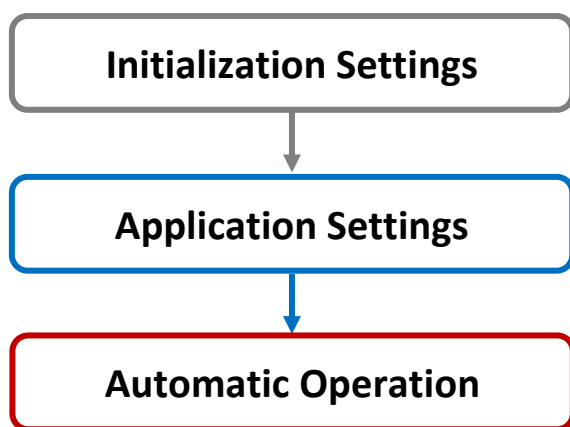


When using KW007 in **I²C mode**,
the IC pin 19 (Range1) must connect to GND.
(or pull low by MCU or 0R resistor to GND)

KW007 Doppler Radar Software Procedure

The **KW007** Doppler radar sensor can operate via a standard **I²C interface**. Upon power-up, the device must perform the following initialization sequence to ensure stable operation and optimal detection performance.

1. **Initialization Settings:** Write the fundamental chip configuration parameters to initialize the device.
2. **Application Settings:** Configure the sensing parameters based on specific application requirements. This includes setting the detection range, and Doppler filter corners.
3. **Automatic Operation:** Once the configuration is applied, the radar operates automatically. The host MCU periodically reads the radar parameters at set intervals for subsequent application processing.



Standard I²C Write/Read Interface Description for KW007

Supported Modes: Standard-mode (100 kHz) / Fast-mode (400 kHz)

Device Address: 7-bit addressing

Data Format: 8-bit Data, MSB First

KW007 Write/Read Address

The KW007 uses an addressing scheme consisting of a fixed 7-bit slave address ID followed by a read/write (R/W) bit.

◆ **Slave address ID:** The fixed 8-bit slave address ID is **00110010b (0x32h)**.

◆ **Read/Write Control:** The address is followed by a R/W bit:

➤ '0' for **Write** operations. (8bit: 0011001**0**b, 0x32h)

➤ '1' for **Read** operations. (8bit: 0011001**1**b, 0x33h)

KW007 Write mode Write format is shown as follow:

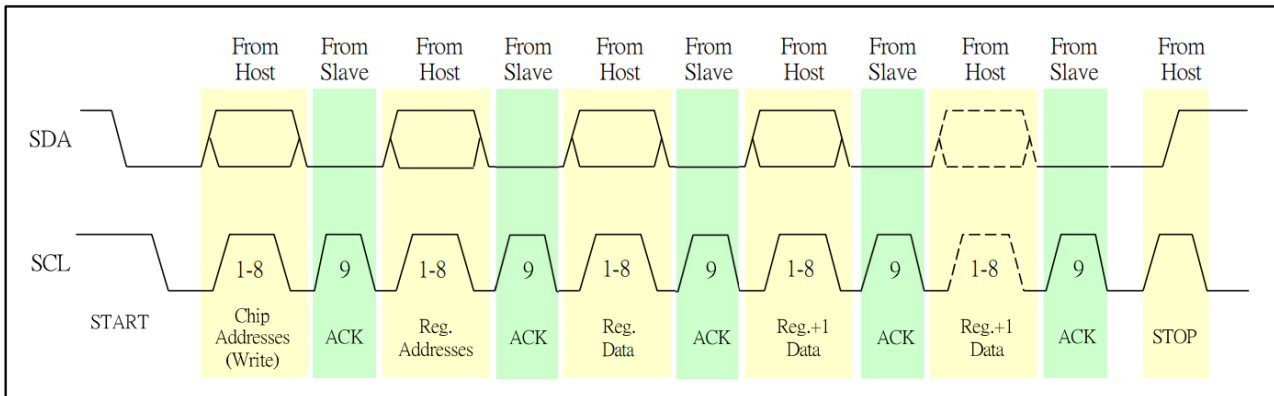


Figure 1. KW007 I²C Write Mode Timing Diagram

KW007 Read mode Read format is shown as follow:

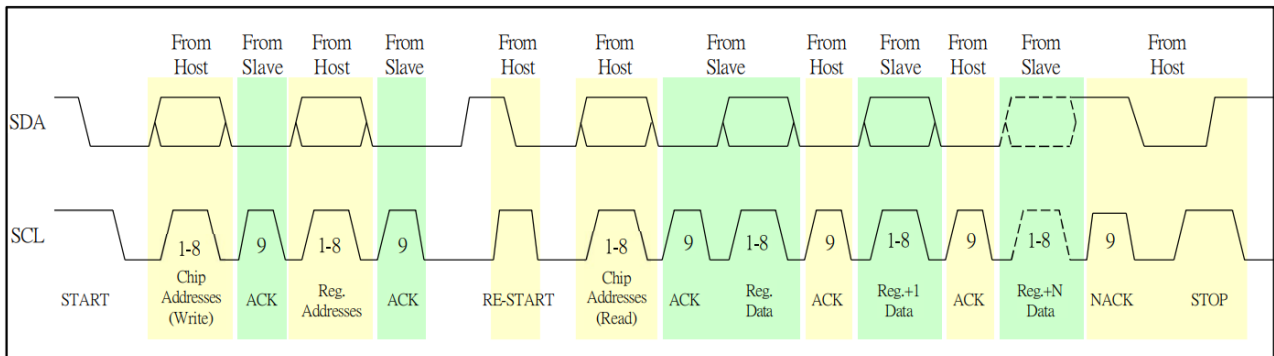


Figure 2. KW007 I²C Read Mode Timing Diagram

Register Descriptions

Register	Function	Access	Value	Bits	Description	
Reg 0	R0	R	-	[7:0]	Reserved	
Reg 1	VCO Bank	R	-	[5:0]	VCO Bank (Read Only, factory burned)	
Reg 2 Reg 4	R2 R4	R	-	[7:0]	Reserved	
Reg 5	RSSI	R	-	[7:0]	RSSI Output (Read Only, 0-255)	
Reg 6	Detection Flag: Day/Night & Approach & Leaving	R	-	[3]	Day/Night Flag 0: Night 1: Day	
				[2]	Flag Forward/Approach 0: No object approaching 1: Object approaching	
				[1]	Flag Backward/Leaving 0: No object leaving 1: Object leaving	
Reg 7	R7	R	-		Reserved	
Reg 8	R8	W/R	0xFC	[7:0]	Reserved	
Reg 9	Manual Mode	W/R	0x30	[5]	[5]: Efuse mode 0: FUSE 1: I2C	
Reg 10	R10	W/R	0x88	[7:0]	Reserved	
Reg 11	R11	W/R	0x03	[7:0]	Reserved	
Reg 12	Sampling Rate	W/R	0x54	[7]	Sampling Rate: 0=1kHz, 1=500Hz	
		W/R		[5:0]	VCO Bank (Set during initialization)	
Reg 13 Reg 18	R13 R18	W/R	-	[7:0]	Reserved	
Reg 19	Amp1_Gain	W/R	0xEA	[7:5]	Amplifier 1 Gain: 000: 8 db 001: 12 db 010: 16 db 011: 19 db 100: 21 db 101: 24 db 110: 28 db 111: 32 db	Total Gain = Amp1_Gain + Amp2_Gain + Amp3_Gain (Ex. 24+24+0 = 48db)
Reg 20	Amp3_Gain	W/R	0xF7	[4:3]	Amplifier 3 Gain: 00: 0 db 01: 4 db 10: 8 db 11: 12 db	
	Amp2_Gain	W/R		[2:0]	Amplifier 2 Gain: 000: 8 db 001: 12 db 010: 16 db 011: 19 db 100: 21 db 101: 24 db 110: 28 db 111: 32 db	

Register	Function	Access	Value	Bits	Description
Reg 21	Slow Speed Detection	W/R	0x5F	[5]	High-pass Corner: (Default = 0) 0: Low (Allows slow-moving signals.) 1: High (Cuts off slow-moving signals.)
Reg 22 Reg 23	R22 R23	W/R	-	[7:0]	Reserved
Reg 24	Enhanced Mode	W/R	0xA0	[6]	Enhanced Range Mode: 0: Off (Normal mode) 1: On (Enhanced mode)
Reg 25 Reg 26	R25 R26	W/R	-	[7:0]	Reserved
Reg 27	Valid Decision Magnitude Threshold	W/R	0x30	[7:0]	Decision Magnitude Threshold 00000000: 0 00000001: 1 00000010: 2 ... 11111111: 255
Reg 28	R28	W/R	0x55	[7:0]	Reserved
Reg 29	GPO Delay	W/R	0x75	[2:1]	GPO Delay: (Default = 10) SE/L S 00: 16s 4s 01: 32s 2s 10: 1s 1s 11: 64s 64s
	High Speed Detection	W/R		[0]	Low-pass Corner: (Default = 1) 0: Low (Filters out ultra-fast signals.) 1: High (allow more fast objects detect.)
Reg 30 Reg 31	R30 R31	W/R	-	[7:0]	Reserved

KW007 Recommended Register Configuration:

Address	Value (Hex)	Address	Value (Hex)	Address	Value (Hex)	Address	Value (Hex)
Reg 0	0x00	Reg 8	0xFC	Reg 16	0x98	Reg 24	0xA0
Reg 1	0x00	Reg 9	0x30	Reg 17	0xA0	Reg 25	0x20
Reg 2	0x00	Reg 10	0x88	Reg 18	0x00	Reg 26	0x9A
Reg 3	0x00	Reg 11	0x03	Reg 19	0xEA	Reg 27	0x30
Reg 4	0x00	Reg 12	0x54	Reg 20	0xF7	Reg 28	0x55
Reg 5	0x00	Reg 13	0x4B	Reg 21	0x5F	Reg 29	0x75
Reg 6	0x00	Reg 14	0x68	Reg 22	0x02	Reg 30	0x42
Reg 7	0x00	Reg 15	0x55	Reg 23	0x30	Reg 31	0x00

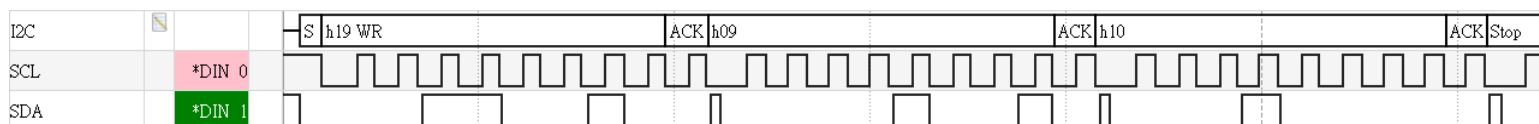
Note: Reg9 = 0x30 (manual mode), Reg12[5:0] will be set to VCO bank value.

Initialization Flow

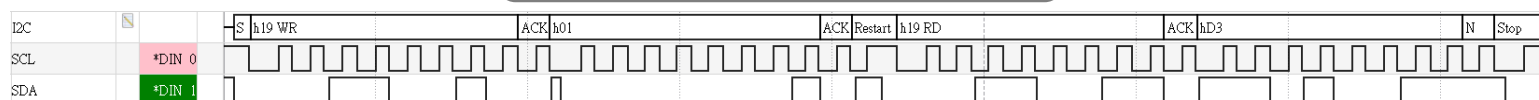
Follow these steps in order:

1. **Write REG9 = 0x10** (Efuse mode) to enable reading the factory-burned VCO bank value
2. **Read REG1[5:0]** to obtain the VCO bank value (factory calibrated for each chip)
3. **Set REG12[5:0] = VCO bank value** in your initialization register array
4. **Write all 32 registers (R0-R31)** with REG9 = 0x30 (I²C manual mode)

Write Reg9 = 0x10

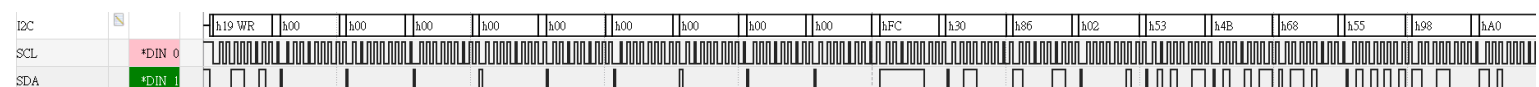


Read Reg1[5:0]



Set Reg12[5:0] = VCO bank

Write all 32 Registers



1. Functional Description

The detection range is determined by the three-stage amplifier gain (AmpX_Gain), the decision magnitude threshold. Increasing the amplifier gain and reducing the magnitude threshold will extend the detection range; however, these settings also increase susceptibility to false triggers.

Three-stage Amplifier Gain:

Amp1_Gain R19[7:5]:

(000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

Amp2_Gain R20[2:0]:

(000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

Amp3_Gain R20[4:3]:

(000=0dB, 001=4dB, 010=8dB, 011=12dB)

For example, to set gain to 48 dB:

Write amp1_gain 32dB, write amp2_gain 16dB, write amp3_gain 0dB.

Total gain:

$$32 + 16 + 0 = 48 \text{ dB}$$

(We suggested to use higher gain in previous stage for better signal/noise quality.)

Mag Threshold:

Reg27 set the signal threshold after three-stage amplifier, signal need larger than mag threshold to do detection algorithm. (Range 0~255).

Note: If gain too small or too high, the signal strength saturated to 0 or 255 and algorithm not work well.

Enhance Range Mode:

When Reg24[6] = 1, the detection range will increase.

Doppler Filter High-Pass Corner

The Reg21[5] defines the high-pass corner frequency of the Doppler filter.

- 0: Low corner frequency.
Allows low-frequency signals to pass; capable of detecting subtle movements (e.g., breathing).
- 1: High corner frequency.
Cuts off slow-moving signals; optimized for detecting rapid motion and filtering out environmental drifts.

Doppler Filter Low-Pass Corner

The Reg29[0] defines the low-pass corner frequency of the Doppler filter.

- 0: Low corner frequency.
Filters out ultra-fast signals (likely interference) to provide a steady and reliable output.
- 1: High corner frequency.
Best for speed. It allows the radar to "see" everything instantly, making it highly sensitive to objects moving at extreme speeds.

Sampling Rate:

Sampling rate Reg12[7] will influence filter corner.

Ex. $\text{Corner}_{fs(1\text{kHz})} = 2 \times \text{Corner}_{fs(500\text{Hz})}$.

*fs = Sampling Frequency.

The detection object speed is determined by doppler filter **corner** and **sampling rate**.
The Doppler low-pass corner filters out fast-moving clutter, but may also attenuate fast-walking human targets. The Doppler high-pass corner is related to slow-motion clutter rejection.

GPO Delay Time

The Reg29[2:1] defines the delay time for the GPO output.

To prevent the output from turning ON and OFF too quickly (chattering) when a target is at the edge of the detection range.

Once motion is detected, the GPO triggers. When the motion stops, the GPO will wait for the **Delay Time** to expire before returning to the "Idle" state.

- 00: 16 Sec
- 01: 32 Sec
- 10: 1 Sec
- 11: 64 Sec

Approach Flag

The Reg6[2] defines whether an object is moving toward the radar.

Read only.

- 0: No object approaching.
- 1: Object approaching.

Leaving Flag

The Reg6[1] defines whether an object is moving away from the radar.

Read only.

- 0: No object leaving.
- 1: Object leaving.

Day/Night Flag

The Reg6[1] defines ambient light condition.

Read only.

- 0: Night.
- 1: Day.

Recommended Settings Table

Detection Range (Typical Estimate)	Total Gain Reg19[7:5] + Reg20[4:0] = AMP1 + AMP2 + AMP3	Magnitude Threshold Reg27[7:0]	Enhanced Mode Reg24[6]
Level 1 (15 cm)	19+8+0 = 27 dB	45	OFF
Level 2 (50cm)	24+8+0 = 32dB	64	OFF
Level 3 (1 m)	32+12+0 = 44 dB	90	OFF
Level 4 (3 m)	32+21+0 = 53 dB	90	OFF
Level 5 (5 m)	32+32+0 = 64 dB	90	OFF
Level 6 (7 m)	32+32+4 = 68 dB	64	OFF
Level 7 (10 m)	32+32+12 = 76 dB	32	OFF
Level 8 (13 m)	32+32+12= 76 dB	32	ON

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Revision History	
V1.6	Reg29 0x65 -> 0x75
V1.7	Modify Reg9 Description Update Recommended Table (Level 0~7)
V1.8	Modify Settings Table
V1.9	Modify Register Table

End of Document